Manage cluster peers

ONTAP 9.11.1 REST API reference

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Manage cluster peers

Cluster peers endpoint overview

Overview

Cluster peering allows administrators of ONTAP systems to establish relationships between two or more independent clusters. When a relationship exists between two clusters, the clusters can exchange user data and configuration information, and coordinate operations. The /cluster/peers endpoint supports create, get, modify, and delete operations using GET, PATCH, POST and DELETE HTTP requests.

Create a cluster peer

You can set up a new cluster peer relationship by issuing a POST request to /cluster/peers. Parameters in the POST body define the settings of the peering relationship. A successful POST request that succeeds in creating a peer returns HTTP status code "201", along with the details of the created peer, such as peer UUID, name, and authentication information. A failed POST request returns an HTTP error code along with a message indicating the reason for the error. This can include malformed requests and invalid operations.

Examples of creating cluster peers

Creating a cluster peer request with an empty request to accept the defaults

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "86de6c46-bdad-11eb-83cd-005056bb267e",
    "name": "Clus_fghf",
    "authentication": {
      "passphrase": "pLznaom1ctesJFq4kt5Qfghf",
      "expiry_time": "2021-05-25T20:04:15-04:00"
    },
    "ip_address": "0.0.0.0",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/86de6c46-bdad-11eb-83cd-005056bb267e"
      }
    }
  }
]
}
```

Creating a cluster peer request with a system-generated passphrase that will expire on 05/26/2021 at 12:34:56

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers' -d '{"authentication":
{"expiry_time": "05/26/2021 12:34:56", "generate_passphrase": true}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "14c817c7-bdad-11eb-83cd-005056bb267e",
    "name": "Clus_F6ht",
    "authentication": {
      "passphrase": "dZNOKkpVfntNZHf3MjpNF6ht",
      "expiry_time": "2021-05-26T12:34:56-04:00"
    },
    "ip_address": "0.0.0.0",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/14c817c7-bdad-11eb-83cd-005056bb267e"
      }
    }
  }
]
}
```

Creating a cluster peer request with a peer address and the generated passphrase is returned in the response

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers' -d '{"remote":
{"ip_addresses": ["1.2.3.4"]}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "b404cc52-bdae-11eb-812c-005056bb0af1",
    "name": "",
    "authentication": {
      "passphrase": "yDhdOteVGEOhkeXF+DJYwDro",
      "expiry_time": "2021-05-25T20:28:12-04:00"
    },
    "_links": {
      "self": {
        "href": "/api/cluster/peers/b404cc52-bdae-11eb-812c-005056bb0af1"
      }
    }
  }
]
}
```

Creating a cluster peer request with a peer name and the generated passphrase is returned in the response

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers' -d '{"name":
"cp_xyz123", "authentication": {"generate_passphrase": true}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "125f8dc6-bdb1-11eb-83cd-005056bb267e",
    "name": "cp_xyz123",
    "authentication": {
      "passphrase": "eeGTerZlh2qSAt2akpYEcM1c",
      "expiry_time": "2021-05-25T20:29:38-04:00"
    },
    "ip_address": "1.2.3.5",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/125f8dc6-bdb1-11eb-83cd-005056bb267e"
      }
    }
  }
]
}
```

Creating a cluster peer request with a name, a peer address, and a passphrase

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers' -d '{"name":
"cp_xyz123", "remote": {"ip_addresses": ["1.2.3.4"]}, "authentication":
{"passphrase": "xyz12345"}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "b404cc52-bdae-11eb-812c-005056bb0af1",
    "authentication": {
      "expiry_time": "2021-05-25T20:32:49-04:00"
    },
    "_links": {
      "self": {
        "href": "/api/cluster/peers/b404cc52-bdae-11eb-812c-005056bb0af1"
      }
    }
  }
]
}
```

Creating a cluster peer request with a proposed encryption protocol

```
# The API:
/api/cluster/peers

# The call:
curl -X POST 'https://<mgmt-ip>/api/cluster/peers' -d '{"encryption":
{"proposed": "tls-psk"}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "b33a23a6-bdb1-11eb-83cd-005056bb267e",
    "name": "Clus_Pslc",
    "authentication": {
      "passphrase": "Gy8SqsXVhcUkS1AfepH7Pslc",
      "expiry_time": "2021-05-25T20:34:07-04:00"
    },
    "ip_address": "1.2.3.5",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/b33a23a6-bdb1-11eb-83cd-005056bb267e"
      }
    }
  }
]
}
```

Creating local intercluster LIFs

The local cluster must have an intercluster LIF on each node for the correct operation of cluster peering. If no local intercluster LIFs exist, you can optionally specify LIFs to be created for each node in the local cluster. These local interfaces, if specified, are created on each node before proceeding with the creation of the cluster peering relationship. Cluster peering relationships are not established if there is an error preventing the LIFs from being created. After local interfaces have been created, do not specify them for subsequent cluster peering relationships.

Local LIF creation fields

- local_network.ip_addresses - List of IP addresses to assign, one per node in the local cluster.
- local_network.netmask - IPv4 mask or subnet mask length.
- local_network.broadcast_domain - Broadcast domain that is in use within the IPspace.
- local_network.gateway - The IPv4 or IPv6 address of the default router.

Additional information on network routes

When creating LIFs, the network route discovery mechanism might take additional time (1-5 seconds) to become visible in the network outside of the cluster. This delay in publishing the routes might cause an initial cluster peer "create" request to fail. This error disappears with a retry of the same request.

This example shows the POST body when creating four intercluster LIFs on a 4-node cluster before creating a cluster peer relationship.

```
# The API:
/api/cluster/peers

# The call:
cluster_peer_4_node.txt:
{
  "local_network":
  {
      "interfaces": [
          {"ip_address":"1.2.3.4"},
          {"ip_address":"1.2.3.5"},
          {"ip_address":"1.2.3.6"}
      ],
      "netmask": "255.255.0.0",
      "broadcast_domain": "Default",
      "gateway": "1.2.0.1"
  },
  "remote": {"ip_addresses": ["1.2.9.9"]},
  "authentication": {"passphrase": "xyz12345"}
}
curl -X POST "https://<mgmt-ip>/api/cluster/peers" -d
"@cluster_peer_4_node.txt"

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "b404cc52-bdae-11eb-812c-005056bb0af1",
    "local_network": {
      "interfaces": [
        {
          "ip_address": "1.2.3.4"
        },
        {
          "ip_address": "1.2.3.5"
        },
        {
```

```
          "ip_address": "1.2.3.6"
        }
      ]
    },
    "authentication": {
      "expiry_time": "2021-05-25T21:28:26-04:00"
    },
    "_links": {
      "self": {
        "href": "/api/cluster/peers/b404cc52-bdae-11eb-812c-005056bb0af1"
      }
    }
  }
]
}
```


Examples of retrieving existing cluster peers

You can retrieve peers in a cluster by issuing a GET request to /cluster/peers. It is also possible to retrieve a specific peer when qualified by its UUID to /cluster/peers/{uuid}. A GET request might have no query parameters or a valid cluster UUID. The former retrieves all records while the latter retrieves the record for the cluster peer with that UUID.

Retrieving all cluster peer relationships, both established and pending

```
# The API:
/api/cluster/peers

# The call:
curl 'https://<mgmt-ip>/api/cluster/peers'

# The response:
{
"records": [
  {
    "uuid": "a6001076-bdb2-11eb-83cd-005056bb267e",
    "name": "Clus_bH6l",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/a6001076-bdb2-11eb-83cd-005056bb267e"
      },
      "interfaces": {
        "href":
"/api/network/ip/interfaces?services=intercluster_core&ipspace.uuid=0bac5c
ed-a911-11eb-83cd-005056bb267e"
      }
    }
  },
  {
    "uuid": "b404cc52-bdae-11eb-812c-005056bb0af1",
    "name": "remote-cluster",
    "_links": {
      "self": {
        "href": "/api/cluster/peers/b404cc52-bdae-11eb-812c-005056bb0af1"
      },
      "interfaces": {
        "href":
"/api/network/ip/interfaces?services=intercluster_core&ipspace.uuid=0bac5c
ed-a911-11eb-83cd-005056bb267e"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/cluster/peers"
  }
}
}
```

Retrieving all cluster peer relationships which are not in an available state

```
# The API:
/api/cluster/peers

# The call:
curl 'https://<mgmt-ip>/api/cluster/peers?status.state=!available'

# The response:
{
"records": [
  {
    "uuid": "a6001076-bdb2-11eb-83cd-005056bb267e",
    "name": "Clus_bH6l",
    "status": {
      "state": "unidentified"
    },
    "_links": {
      "self": {
        "href": "/api/cluster/peers/a6001076-bdb2-11eb-83cd-005056bb267e"
      },
      "interfaces": {
        "href":
"/api/network/ip/interfaces?services=intercluster_core&ipspace.uuid=0bac5c
ed-a911-11eb-83cd-005056bb267e"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/cluster/peers?status.state=!available"
  }
}
}
```

Retrieving information about a single cluster peer relationship

```
# The API:
/api/cluster/peers

# The call:
curl 'https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-812c-
005056bb0af1'
```

```
# The response:
{
"uuid": "b404cc52-bdae-11eb-812c-005056bb0af1",
"name": "remote-cluster",
"version": {
  "full": "NetApp Release Stormking__9.10.1: Tue May 25 08:08:44 UTC
2021",
  "generation": 9,
  "major": 10,
  "minor": 1
},
"status": {
  "state": "available",
  "update_time": "2021-05-25T19:38:55-04:00"
},
"ipspace": {
  "uuid": "0bac5ced-a911-11eb-83cd-005056bb267e",
  "name": "Default",
  "_links": {
    "self": {
      "href": "/api/network/ipspaces/0bac5ced-a911-11eb-83cd-005056bb267e"
    }
  }
},
"remote": {
  "name": "remote-cluster",
  "serial_number": "1-80-000011",
  "ip_addresses": [
    "1.2.3.4"
  ]
},
"authentication": {
  "in_use": "ok",
  "state": "ok"
},
"encryption": {
  "state": "tls_psk"
},
"_links": {
  "self": {
    "href": "/api/cluster/peers/b404cc52-bdae-11eb-812c-005056bb0af1"
  },
  "interfaces": {
    "href":
"/api/network/ip/interfaces?services=intercluster_core&ipspace.uuid=0bac5c
```

```
ed-a911-11eb-83cd-005056bb267e"
  }
}
}
```

Examples of updating an existing cluster peer

You can update a cluster peer relationship by issuing a PATCH request to /cluster/peers/{uuid}. As in the CLI mode, you can toggle the proposed encryption protocol, update the passphrase, or specify a new set of stable addresses. All PATCH requests take the parameters that are to be updated in the request body. If generate_passphrase is "true", the passphrase is returned in the PATCH response.

Updating the proposed encryption protocol from tls-psk to none

```
# The API:
/api/cluster/peers

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-
812c-005056bb0af1' -d '{"authentication": {"passphrase": "xyz12345",
"in_use": "ok"}, "encryption": {"proposed": "none"}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "authentication": {
      "passphrase": "xyz12345",
      "in_use": "ok"
    },
    "encryption": {
      "proposed": "none"
    }
  }
]
}
```

Updating the passphrase

```
# The API:
/api/cluster/peers

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-
812c-005056bb0af1' -d '{"authentication": {"passphrase": "xyz12345",
"in_use": "ok"}}'

# The response:
{
"num_records": 1,
"records": [
  {
    "authentication": {
      "passphrase": "xyz12345",
      "in_use": "ok"
    }
  }
]
}
```

Setting an auto-generated passphrase

```
# The API:
/api/cluster/peers

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-
812c-005056bb0af1' -d '{"authentication": {"generate_passphrase": true,
"in_use": "ok"}}'

# The response:
{}
```

Updating remote IP addresses

```
# The API:
/api/cluster/peers

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-
812c-005056bb0af1' -d '{"remote": {"ip_addresses": ["1.2.3.6"]}}'

# The response:
{}
```

An example of deleting an existing cluster peer

You can delete a cluster peer using the HTTP DELETE request.

Deleting a peer with peer UUID "8becc0d4-c12c-11e8-9ceb-005056bbd143"

```
# The API:
/api/cluster/peers

# The call:
curl -X DELETE "https://<mgmt-ip>/api/cluster/peers/b404cc52-bdae-11eb-
812c-005056bb0af1"

# The response:
{}
```

Retrieve cluster peers

GET `/cluster/peers`

Introduced In: 9.6

Retrieves the collection of cluster peers.

Parameters

Name	Type	In	Required	Description
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[cluster_peer]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "interfaces": {
        "href": "/api/resourcelink"
      },
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "authentication": {
      "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
      "in_use": "ok",
      "state": "ok"
    },
    "encryption": {
      "proposed": "none",
      "state": "none"
    },
    "initial_allowed_svms": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "ipspace": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "exchange",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
```

```
      },
      "local_network": {
        "broadcast_domain": "bd1",
        "gateway": "10.1.1.1",
        "interfaces": {
          "ip_address": "10.10.10.7"
        },
        "netmask": "255.255.0.0"
      },
      "name": "cluster2",
      "peer_applications": [
        "snapmirror",
        "flexcache"
      ],
      "remote": {
        "ip_addresses": {
        },
        "name": "cluster2",
        "serial_number": "4048820-60-9"
      },
      "status": {
        "state": "available",
        "update_time": "2017-01-25T11:20:13Z"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
      "version": {
        "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
        "generation": 9,
        "major": 4,
        "minor": 0,
        "patch": "P2"
      }
    }
  }
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
interfaces	href	
self	href	

authentication

Name	Type	Description
expiry_time	string	The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour.
generate_passphrase	boolean	Auto generate a passphrase when true.
in_use	string	
passphrase	string	A password to authenticate the cluster peer relationship.
state	string	

encryption

Name	Type	Description
proposed	string	
state	string	

_links

Name	Type	Description
self	href	

initial_allowed_svms

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipspace

The IPspace of the local intercluster LIFs.

Name	Type	Description
_links	_links	
name	string	IPspace name
uuid	string	IPspace UUID

interfaces

Name	Type	Description
ip_address	string	IPv4 or IPv6 address

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

Name	Type	Description
broadcast_domain	string	Broadcast domain that is in use within the IPspace.
gateway	string	The IPv4 or IPv6 address of the default router.
interfaces	array[interfaces]	
netmask	string	IPv4 mask or netmask length.

remote

Name	Type	Description
ip_addresses	array[string]	The IPv4 addresses, IPv6 addresses, or hostnames of the peers.
name	string	The name of the remote cluster.
serial_number	string	The serial number of the remote cluster.

status

Name	Type	Description
state	string	
update_time	string	The last time the state was updated.

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Name	Type	Description
full	string	The full cluster version string.
generation	integer	The generation portion of the version.
major	integer	The major portion of the version.
minor	integer	The minor portion of the version.
patch	string	The patch portion of the version.

cluster_peer

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	

Name	Type	Description
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create a peering relationship

POST `/cluster/peers`

Introduced In: 9.6

Creates a peering relationship and, optionally, the IP interfaces it will use. There are two methods used to create a peering relationship:

- Provide a remote IP address - Used when creating a new cluster peer relationship with a specific remote cluster. This requires at least one remote intercluster IP address from the remote cluster.
- Do not provide a remote IP address - Used when the remote IP address is not provided and when the storage system is ready to accept peering requests from foreign clusters.

Required properties

- `remote.ip_addresses` - Addresses of the remote peers. The local peer must be able to reach and connect to these addresses for the request to succeed in creating a peer. Only required when creating a peering relationship by providing a remote IP address.
- Either set `generate_passphrase` to "true" or provide a passphrase in the body of the request. Only one of these options is required.

Recommended optional properties

- `name` - Name of the peering relationship or name of the remote peer.
- `passphrase` - User generated passphrase for use in authentication.
- `generate_passphrase` (true/false) - When "true", ONTAP automatically generates a passphrase to authenticate cluster peers.
- `ipspace` - IPspace of the local intercluster LIFs. Assumes Default IPspace if not provided.
- `initial_allowed_svms` - Local SVMs allowed to peer with the peer cluster's SVMs. Can be modified until the remote cluster accepts this cluster peering relationship.
- `local_network` - Fields to create a local intercluster LIF.

- `expiry_time` - Duration in ISO 8601 format for which the user-supplied or auto-generated passphrase is valid. Expiration time must not be greater than seven days into the future. ISO 8601 duration format is "PnDTnHnMnS" or "PnW" where n is a positive integer. The "nD", "nH", "nM" and "nS" fields can be dropped if zero. "P" must always be present and "T" must be present if there are any hours, minutes, or seconds fields.
- `encryption_proposed` (none/tls-psk) - Encryption mechanism of the communication channel between the two peers.
- `peer_applications` - SVM peering applications (SnapMirror, FlexCache or both) for which the SVM peering relationship is set up.

Additional information

As with creating a cluster peer through the CLI, the combinations of options must be valid in order for the create operation to succeed. The following list shows the combinations that will succeed and those that will fail:

- A passphrase only (fail)
- A peer IP address (fail)
- A passphrase with an expiration time > 7 days into the future (fail)
- A peer IP address and a passphrase (OK)
- generate_passphrase=true (OK)
- Any proposed encryption protocol (OK)
- An IPspace name or UUID (OK)
- A passphrase, peer IP address, and any proposed encryption protocol (OK)
- A non empty list of initial allowed SVM peer names or UUIDs. (OK)

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	

Name	Type	Description
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Example request


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
    "name": "cluster2",
    "peer_applications": [
      "snapmirror",
      "flexcache"
    ],
    "remote": {
      "ip_addresses": {
      },
      "name": "cluster2",
      "serial_number": "4048820-60-9"
    },
    "status": {
      "state": "available",
      "update_time": "2017-01-25T11:20:13Z"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "version": {
      "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
      "generation": 9,
      "major": 4,
      "minor": 0,
      "patch": "P2"
    }
  }
```


Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
authentication	authentication	
ip_address	string	IPv4 or IPv6 address
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster, or a temporary name might be autogenerated for anonymous cluster peer offers.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "2017-01-25T11:20:13Z"
  },
  "ip_address": "10.10.10.7",
  "name": "cluster2"
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
1966366	The system SVM of the cluster IPspace hosts cluster LIFs only.
4653365	IPspaces are unavailable with cluster peering: {ipspace}.
4656069	Specifying a passphrase without remote IP addresses is not supported.
4656070	The encryption protocol is meaningful only with authenticated cluster peer relationships.
4656071	Cannot peer with a cluster bearing the same name as the local cluster.
4656072	The name must conform to the same rules as a cluster name.
4656074	Cannot check whether all nodes of this cluster support encryption.
4656075	Cannot specify encryption: this operation requires an ECV of 9.6.0 or later.
4656077	Specify either remote IP addresses or generate_passphrase.

Error Code	Description
4656079	No cluster nodes were found. Check your cluster configuration.
4656081	Creating an intercluster LIF requires a list of local IP addresses.
4656085	Cannot create an intercluster LIF with an empty list of local IP addresses.
4656086	Creating an intercluster LIF requires a broadcast domain that is in use within the IPspace.
4656087	The number of local intercluster IP addresses must be less than or equal to the number of available nodes.
4656088	Found no ports matching the IPspace and the broadcast domain.
4656089	Found no matching entry for IPspace.
4656090	The given IPspace differs from the IPspace entry found.
4656091	Creating an intercluster LIF requires a subnet mask or a subnet mask length.
4656096	Creating an intercluster LIF requires an IPv4 or IPv6 address of the default router.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
interfaces	href	
self	href	

authentication

Name	Type	Description
expiry_time	string	The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour.
generate_passphrase	boolean	Auto generate a passphrase when true.
in_use	string	
passphrase	string	A password to authenticate the cluster peer relationship.
state	string	

encryption

Name	Type	Description
proposed	string	
state	string	

_links

Name	Type	Description
self	href	

initial_allowed_svms

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipspace

The IPspace of the local intercluster LIFs.

Name	Type	Description
_links	_links	
name	string	IPspace name
uuid	string	IPspace UUID

interfaces

Name	Type	Description
ip_address	string	IPv4 or IPv6 address

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

Name	Type	Description
broadcast_domain	string	Broadcast domain that is in use within the IPspace.
gateway	string	The IPv4 or IPv6 address of the default router.
interfaces	array[interfaces]	
netmask	string	IPv4 mask or netmask length.

remote

Name	Type	Description
ip_addresses	array[string]	The IPv4 addresses, IPv6 addresses, or hostnames of the peers.

Name	Type	Description
name	string	The name of the remote cluster.
serial_number	string	The serial number of the remote cluster.

status

Name	Type	Description
state	string	
update_time	string	The last time the state was updated.

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Name	Type	Description
full	string	The full cluster version string.
generation	integer	The generation portion of the version.
major	integer	The major portion of the version.
minor	integer	The minor portion of the version.
patch	string	The patch portion of the version.

cluster_peer

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.

Name	Type	Description
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

authentication

Name	Type	Description
expiry_time	string	The date and time the passphrase will expire. The default expiry time is one hour.
passphrase	string	A password to authenticate the cluster peer relationship.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Delete a cluster peer

DELETE `/cluster/peers/{uuid}`

Introduced In: 9.6

Deletes a cluster peer.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	Cluster peer relationship UUID

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
4663070	Unable to delete cluster peer relationship due to an ongoing Vserver migration.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve a cluster peer instance

GET `/cluster/peers/{uuid}`

Introduced In: 9.6

Retrieves a specific cluster peer instance.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	Cluster peer relationship UUID
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Example response


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
    "name": "cluster2",
    "peer_applications": [
      "snapmirror",
      "flexcache"
    ],
    "remote": {
      "ip_addresses": {
      },
      "name": "cluster2",
      "serial_number": "4048820-60-9"
    },
    "status": {
      "state": "available",
      "update_time": "2017-01-25T11:20:13Z"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "version": {
      "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
      "generation": 9,
      "major": 4,
      "minor": 0,
      "patch": "P2"
    }
  }
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
interfaces	href	
self	href	

authentication

Name	Type	Description
expiry_time	string	The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour.
generate_passphrase	boolean	Auto generate a passphrase when true.
in_use	string	
passphrase	string	A password to authenticate the cluster peer relationship.
state	string	

encryption

Name	Type	Description
proposed	string	
state	string	

_links

Name	Type	Description
self	href	

initial_allowed_svms

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipspace

The IPspace of the local intercluster LIFs.

Name	Type	Description
_links	_links	
name	string	IPspace name
uuid	string	IPspace UUID

interfaces

Name	Type	Description
ip_address	string	IPv4 or IPv6 address

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

Name	Type	Description
broadcast_domain	string	Broadcast domain that is in use within the IPspace.
gateway	string	The IPv4 or IPv6 address of the default router.
interfaces	array[interfaces]	
netmask	string	IPv4 mask or netmask length.

remote

Name	Type	Description
ip_addresses	array[string]	The IPv4 addresses, IPv6 addresses, or hostnames of the peers.

Name	Type	Description
name	string	The name of the remote cluster.
serial_number	string	The serial number of the remote cluster.

status

Name	Type	Description
state	string	
update_time	string	The last time the state was updated.

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Name	Type	Description
full	string	The full cluster version string.
generation	integer	The generation portion of the version.
major	integer	The major portion of the version.
minor	integer	The minor portion of the version.
patch	string	The patch portion of the version.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments

Name	Type	Description
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update a cluster peer instance

PATCH `/cluster/peers/{uuid}`

Introduced In: 9.6

Updates a cluster peer instance.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	Cluster peer relationship UUID

Request Body

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

Name	Type	Description
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Example request


```
{
  "_links": {
    "interfaces": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "P1DT2H3M4S or '2017-01-25T11:20:13Z'",
    "in_use": "ok",
    "state": "ok"
  },
  "encryption": {
    "proposed": "none",
    "state": "none"
  },
  "initial_allowed_svms": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "local_network": {
    "broadcast_domain": "bd1",
    "gateway": "10.1.1.1",
    "interfaces": {
      "ip_address": "10.10.10.7"
    },
    "netmask": "255.255.0.0"
  },
```

```
    "name": "cluster2",
    "peer_applications": [
      "snapmirror",
      "flexcache"
    ],
    "remote": {
      "ip_addresses": {
      },
      "name": "cluster2",
      "serial_number": "4048820-60-9"
    },
    "status": {
      "state": "available",
      "update_time": "2017-01-25T11:20:13Z"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "version": {
      "full": "NetApp Release 9.4.0: Sun Nov 05 18:20:57 UTC 2017",
      "generation": 9,
      "major": 4,
      "minor": 0,
      "patch": "P2"
    }
  }
```


Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
authentication	authentication	
ip_address	string	IPv4 or IPv6 address
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster, or a temporary name might be autogenerated for anonymous cluster peer offers.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication": {
    "expiry_time": "2017-01-25T11:20:13Z"
  },
  "ip_address": "10.10.10.7",
  "name": "cluster2"
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
4653261	Error finding IPspace.
4655058	Expiration time cannot be more than 7 days in the future.
4656070	The encryption protocol is meaningful only with authenticated cluster peer relationships.
4656072	The name must conform to the same rules as a cluster name.
4656073	Changing the encryption state requires the refreshing of the authentication passphrase.
4656075	Cannot specify encryption: this operation requires an ECV of ONTAP 9.6.0 or later.
4656076	Cluster peer modify was attempted with mismatched IPv4 and IPv6 addresses.
4656081	The remote IP address list is empty.
4656082	Specify either a passphrase or "-generate-passphrase".

Error Code	Description
4656083	Cannot auto-generate a passphrase when "generate-passphrase" is false. Modifying a passphrase using an auto-generated passphrase requires "generate-passphrase" be true.
4656084	Passphrase can only be modified with an authenticated cluster peer relationship.
4656092	Cluster peer modify was attempted with a host name that did not resolve to an IPv4 or IPv6 address.
4656095	The address family of the specified peer addresses is not valid in this IPspace. Use /api/network/interfaces/ to verify that required LIFs are present and operational on each cluster node.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
interfaces	href	
self	href	

authentication

Name	Type	Description
expiry_time	string	The time when the passphrase will expire, in ISO 8601 duration format or date and time format. The default is 1 hour.
generate_passphrase	boolean	Auto generate a passphrase when true.
in_use	string	
passphrase	string	A password to authenticate the cluster peer relationship.
state	string	

encryption

Name	Type	Description
proposed	string	
state	string	

_links

Name	Type	Description
self	href	

initial_allowed_svms

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipspace

The IPspace of the local intercluster LIFs.

Name	Type	Description
_links	_links	
name	string	IPspace name
uuid	string	IPspace UUID

interfaces

Name	Type	Description
ip_address	string	IPv4 or IPv6 address

local_network

Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.

Name	Type	Description
broadcast_domain	string	Broadcast domain that is in use within the IPspace.
gateway	string	The IPv4 or IPv6 address of the default router.
interfaces	array[interfaces]	
netmask	string	IPv4 mask or netmask length.

remote

Name	Type	Description
ip_addresses	array[string]	The IPv4 addresses, IPv6 addresses, or hostnames of the peers.

Name	Type	Description
name	string	The name of the remote cluster.
serial_number	string	The serial number of the remote cluster.

status

Name	Type	Description
state	string	
update_time	string	The last time the state was updated.

version

This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

Name	Type	Description
full	string	The full cluster version string.
generation	integer	The generation portion of the version.
major	integer	The major portion of the version.
minor	integer	The minor portion of the version.
patch	string	The patch portion of the version.

cluster_peer

Name	Type	Description
_links	_links	
authentication	authentication	
encryption	encryption	
initial_allowed_svms	array[initial_allowed_svms]	The local SVMs allowed to peer with the peer cluster's SVMs. This list can be modified until the remote cluster accepts this cluster peering relationship.

Name	Type	Description
ipspace	ipspace	The IPspace of the local intercluster LIFs.
local_network	local_network	Cluster peering requires an intercluster LIF on each local node. These can be optionally created by specifying a list of IP addresses corresponding to each node.
name	string	Optional name for the cluster peer relationship. By default, it is the name of the remote cluster.
peer_applications	array[string]	Peering applications against which allowed SVMs are configured.
remote	remote	
status	status	
uuid	string	UUID of the cluster peer relationship. For anonymous cluster peer offers, the UUID will change when the remote cluster accepts the relationship.
version	version	This returns the cluster version information. When the cluster has more than one node, the cluster version is equivalent to the lowest of generation, major, and minor versions on all nodes.

authentication

Name	Type	Description
expiry_time	string	The date and time the passphrase will expire. The default expiry time is one hour.
passphrase	string	A password to authenticate the cluster peer relationship.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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